

POKÉMON

TM

DATA VIZ VERSION



Data Visualization Final Project
S. Calabretta, M. Liotta



ANALYSIS MAIN TOPIC

Answering POKÉMON related questions, to investigate Types and Specs across all generations.

DATA SOURCES

- * All-gen POKÉMON statistics and informations: --> <https://www.kaggle.com/datasets/darkmatternet/ultimate-pokmon-dataset-2025>
- * Preferred POKÉMON (2020) --> <https://pokemon2020.pokemon.com/en-us/>

ITEMS



► User Preferences Vis xl
Type Coherence Vis xl
Params Evolution Vis xl
Contributions xl

FAN FAVORITES POKÉMON DON'T RELY SOLELY ON THEIR (Extreme) POWER

Top 10 most liked Pokémon up to Gen 8 span across middle-to-high base-stat distribution

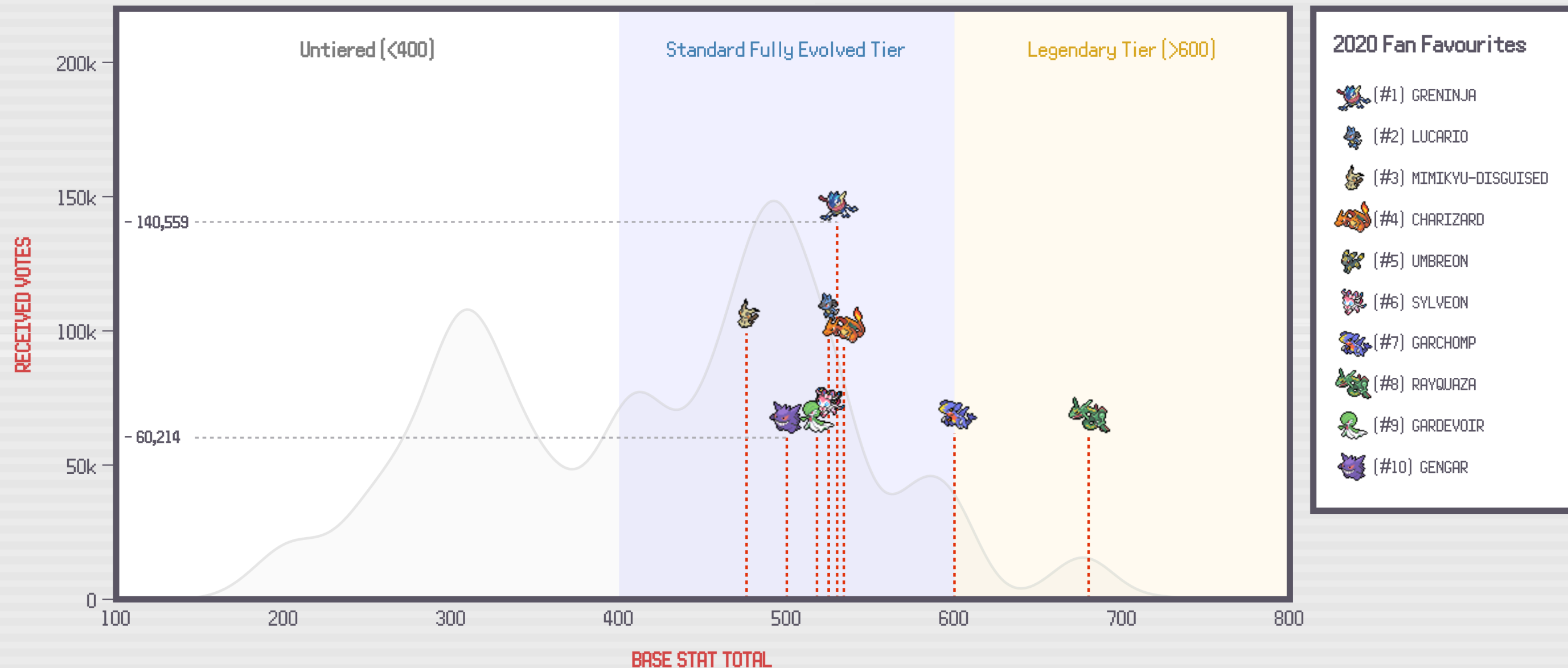


FIGURE. Base Stats of all 905 POKÉMONs up to gen 8. Performed density estimation via KDE, where overlaid POKÉMON sprites resemble in heights the 2020 POKÉMON of the year user preferences. Here showed the 10 highest preferences. Source - <https://pokemon2020.pokemon.com/en-us/>

Data attributes and visual channels

Data Attribute

Attribute Type

Visual Channel

POKéMON user votes

Quantitative

Using X-Axis

POKéMON base stats

Quantitative

Using Y-axis

POKéMON preference
list

Categorical

Via legend and
both min and max
annotations

POKéMON density
having base stats

Quantitative

KDE Density in
the background

ITEMS



User Preferences Vis x0

► Type Coherence Vis x1

Params Evolution Vis x1

Contributions x1

POKÉMON TYPES STAY TRUE TO THEIR NATURE

Comparing the median total stats (Attack + Defense) across all types



FIGURE. Comparison of median total stats (att+def) across all 18 POKÉMON types up to Gen IX, with dashed line representing a balanced 1:1 stats ratio. Highlighted regions show the most heavily skewed types. [kaggle.com/datasets/darkmatternet/ultimate-pokemon-dataset-2025]

Data attributes and visual channels

Data Attribute	Attribute Type	Visual Channel
Median POKÉMON Physical stats per type (att + def)	Quantitative	Using X-axis
Median POKÉMON Special stats per type (att + def)	Quantitative	Using Y-axis
POKÉMON Type	Categorical	Color HUE +symbols
Type combat profile	Categorical	Containment and color HUE

ITEMS



User Preferences Vis x0

Type Coherence Vis x0

► Params Evolution Vis x1

Contributions x1

AN ALMOST THIRTY YEARS LONG PARAMETER TUNING

Evolution of core base statistics alongside specific generation snapshots

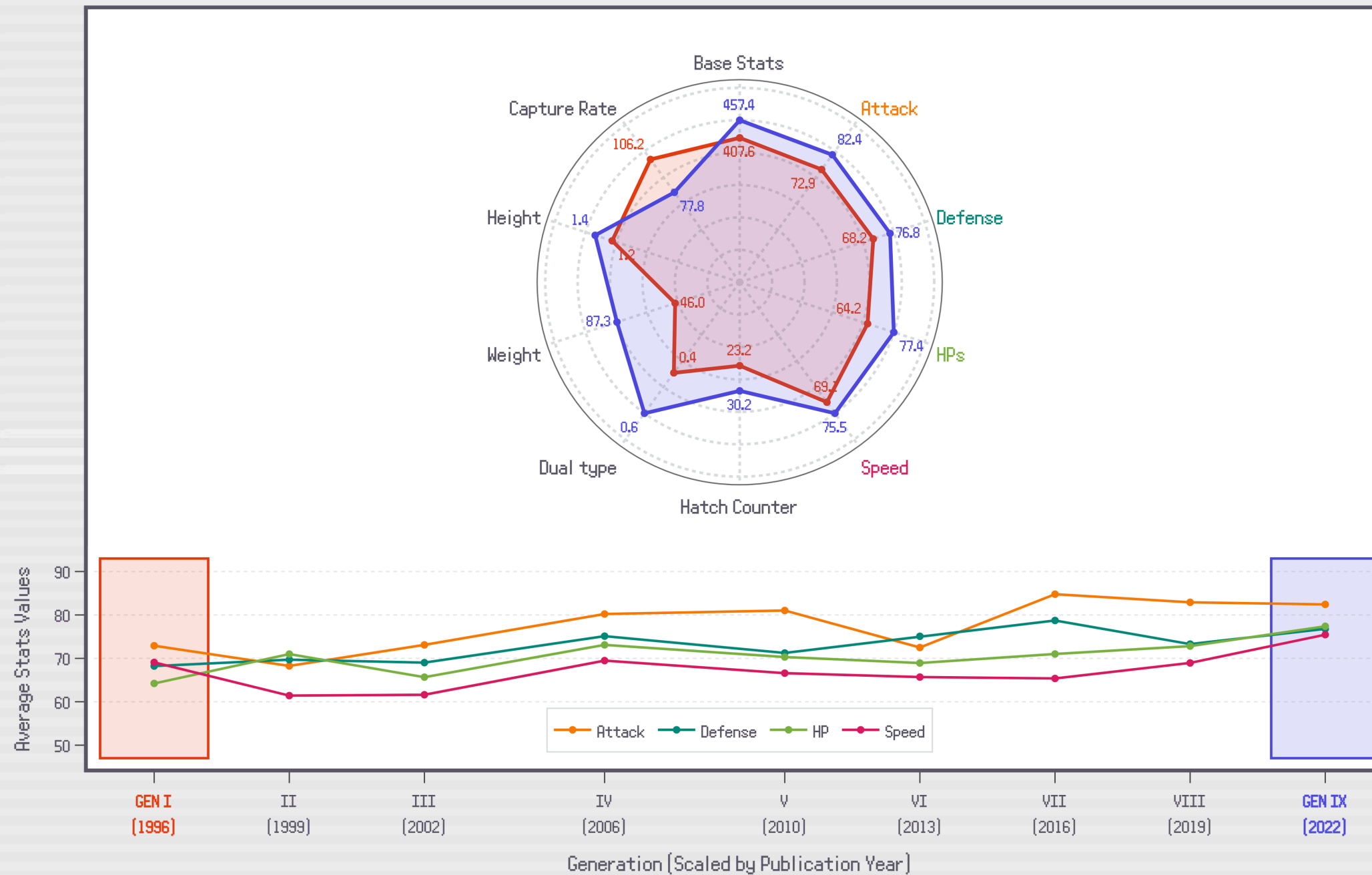


FIGURE. Line chart tracks the true average of HP, Attack, Defense, and Speed across all 9 generations. Above the chart, comprehensive radar profiles compare Gen I and IX averaged statistics. The highlighted boxes emphasize the specific generations showcased in the radar charts.
Source - <https://www.kaggle.com/datasets/darkmatternet/ultimate-pokemon-dataset-2025>.

Data attributes and visual channels

Data Attribute	Attribute Type	Visual Channel
Avg (wrt generation) attack, defense, HPs, Speed, Hatch Counter, num having dual type, Weight, Height and Capture rate	Quantitative	Values on the radius plot, annotated and visually displayed
POKÉMON Generation	Categorical	Different colors and radius
POKÉMON Gen release year	Quantitative	X-Axis spacing

ITEMS



User Preferences Vis x0

Type Coherence Vis x0

Params Evolution Vis x0

► Contributions x1

Project Contributions



Ideas, Data collection and Plot
drafts: Matteo & Silvia

Project Contributions



Visualization one realization:
Matteo

Project Contributions



Visualization two realization:
Silvia

Project Contributions



Visualization three realization:
Matteo & Silvia

Project Contributions



Style and slide design: Matteo &
Silvia

The end



POKéDEX

Discarded Proposal

GREAT POWER COMES WITH STEPS AND POKéBALL CONSUMPTION

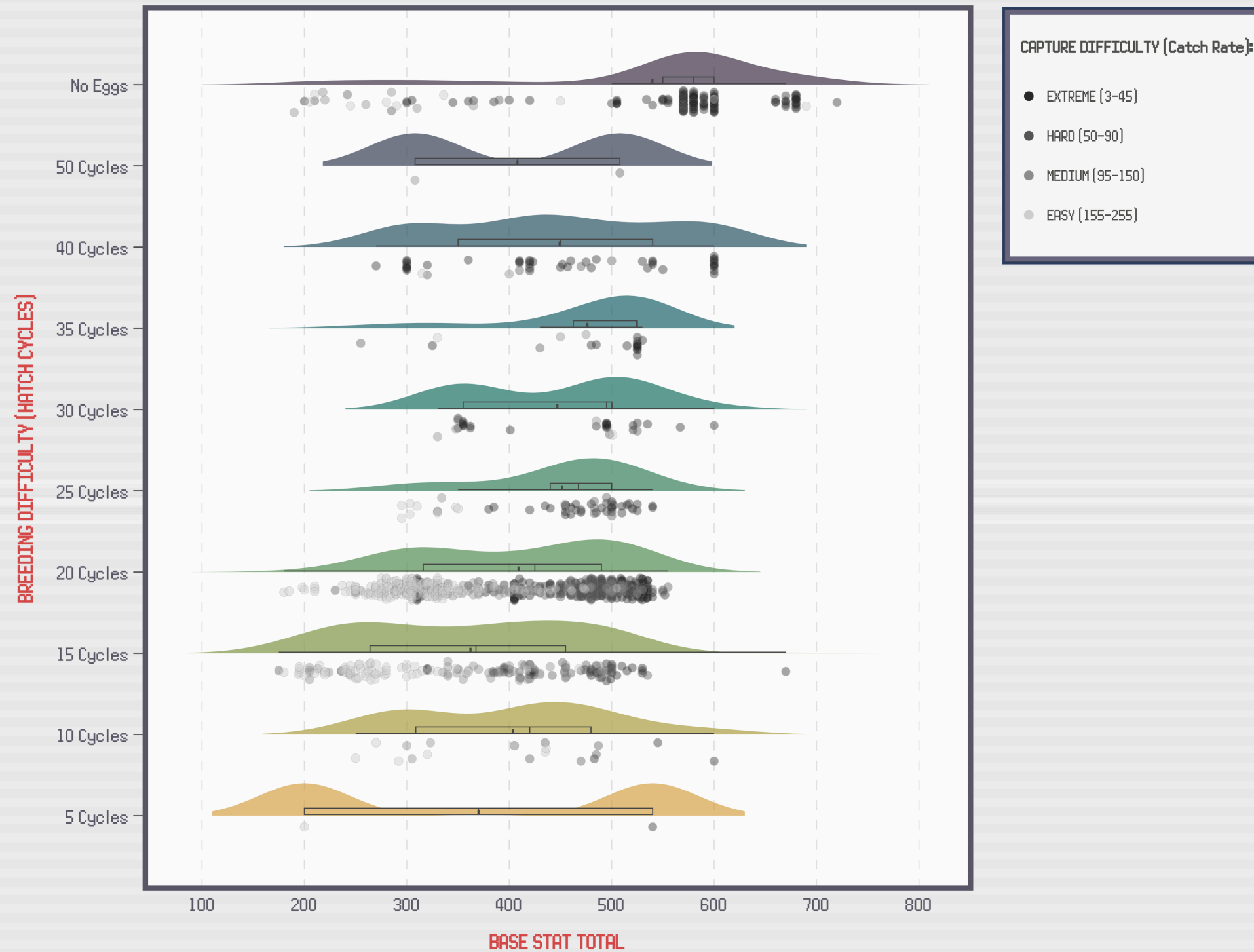


FIGURE. Base Stat POKéMON distribution, grouped by Breeding Difficulty (hatch cycle = 256 steps). While clouds resemble KDE estimation the "rain" resembles individual POKéMONs, colored by Capture Rate. Source - <https://www.kaggle.com/datasets/darkmatternet/ultimate-pokemon-dataset-2025>